Tangents

1 Find the gradient of the tangent to $f(x) = x^3 + 5x$ at the point $(2, 18)$.

2 Consider the function $y = \frac{1}{3}x^3 + \frac{1}{6}x^6 + 2x$.

a Find $\frac{dy}{dx}$.

b Evaluate $\frac{dy}{dx}$ when $x = -7$.

3 Find the gradient of $f(x) = \frac{11}{x} + \frac{10}{x^2}$ at the point $\left(4, \frac{27}{8}\right)$.

4 Consider the parabola $f(x) = x^2 + 3x - 10$.

a Find the x-intercepts.

b Find the gradient of the tangent at the positive x-intercept.

5 Consider the function $y = 4x^2 - 5x + 2$.

a Find $\frac{dy}{dx}$.

b Find the value of x at which the tangent to the parabola is parallel to the x-axis.

6 Consider the tangent to the curve $f(x) = -x^3$ at the point $(2, -8)$.

a Find the gradient of the function $f(x) = -x^3$ at the point.

b Hence, find the equation of the tangent to the curve at the point.

7 Consider the tangent to the curve $f(x) = 5\sqrt{x}$ at the point $\left(\frac{1}{9}, \frac{5}{3}\right)$.

a Find the gradient of the function $f(x) = 5\sqrt{x}$ at the point.

b Hence, find the equation of the tangent to the curve at the point.

8 Consider the tangent to the curve $f(x) = 6\sqrt{x}$ at the point $(4, 12)$.

a Find the gradient of the function $f(x) = 6\sqrt{x}$ at the point.

b Hence, find the equation of the tangent to the curve at the point.

9 Consider the tangent to the curve $f(x) = \frac{2}{x^3}$ at $x = -2$.

a Find the gradient of the function $f(x) = \frac{2}{x^3}$ at $x = -2$.

b Find the y-coordinate of the point of intersection between the tangent line and the curve.

c Hence, find the equation of the tangent to the curve at $x = -2$.

10 Consider the tangent to the curve $f(x) = -2x^2 + 8x + 2$ at $x = 1$.

a Find the y-coordinate of the point of intersection between the tangent line and the curve.

b Find the equation of the tangent to the curve at $x = 1$.

11 Consider the tangent to the curve $f(x) = 3x^3$ at $x = 2$.

a Describe the steps required to find the equation of the tangent line at $x = 2$.

b Hence, find the equation of the tangent to the curve.

12 Consider the function $f(x) = \dfrac{9x + 4}{3x}$.

a Find the y-coordinate of the point on the curve at $x = -1$.

b Hence, find the equation of the tangent to the curve at $x = -1$.

13 Consider the function $g(x) = \dfrac{8x^7 - 6x^6 + 4x^5 + 7}{2x^2}$.

a Find the derivative of the function.

b Find the equation in of the tangent to the curve at $x = 1$.

14 Find the equation of the tangent to $f(x) = x^2 + x$ at the point $(2, 6)$.

15 Find the equation of the tangent to $f(x) = (3x - 1)(2x - 5)$ at the point $(2, -5)$.

16 Consider the function $f(x) = x^3 - 6x^2$. Find the coordinates of the points on the curve where the gradient is 495.

17 Consider the function $f(x) = x^3 + 3x^2 - 19x + 2$. Find the coordinates of the points on the curve where the gradient is 5.

18 Consider the function $f(x) = 2x^2 - 216\sqrt{x}$. Find the coordinates of the point on the curve

19 Consider the function $f\left(x\right)=\frac{4x^3}{3}+\frac{5x^2}{2}-3x+7$. Find the x-coordinates of the points on the curve whose tangent is parallel to the line $y=3x+7$.

20 At point M, the equation of the tangent to the curve $y=x^2$ is given by $y=4x-4$. Find the coordinates of M.

21 At point M, the equation of the tangent to the curve $y=x^3$ is given by $y=12x-16$. Find the coordinates of M.

22 Find the equation of the tangent to the parabola $y=2x^2+8x-5$ at the point where the gradient is 0.

23 Consider the curve given by the function $f\left(x\right)=x^2-4x+2$.

- **a** Find the gradient of the tangent to the curve at the point $(3,-1)$.
- **b** Find the coordinates of the vertex of the parabola $f\left(x\right)$.
- **c** Sketch the graph of the curve and the tangent at the point $(3,-1)$.

24 Consider the curve given by the function $f\left(x\right)=x^2-1$.

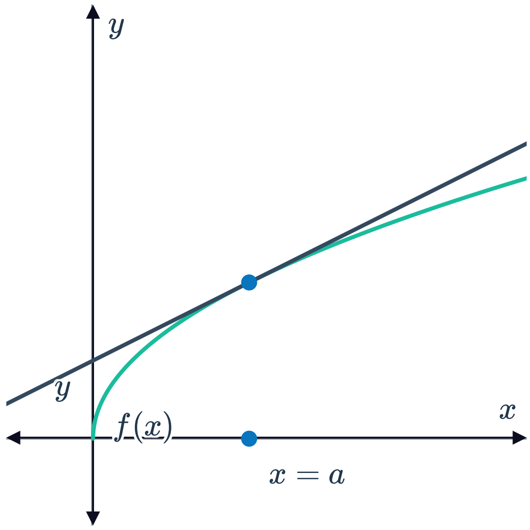
- **a** Find the gradient of the tangent to the curve at the point $(1,0)$.
- **b** Sketch the graph of the curve and the tangent at the point $(1,0)$.

25 Consider the curve $y=x^3-x^2$ and the line $4x-y=11$.

- **a** Find the x-coordinates of the points on the curve at which the tangents are perpendicular to the line $4x-y=11$.
- **b** Find the equation of the tangent to the curve at each of the x-coordinates found in part (a).

26 The curve $f\left(x\right)=k\sqrt{x}-5x$ has a gradient of 0 at $x=16$. Find k.

27 In the figure, the straight line $y = \frac{1}{10}x + b$ [redacted] tangent to the graph of $f(x) = 6\sqrt{x}$ at $x = a$.



a Find a.

b Find b.

28 $5x + y + 2 = 0$ is the tangent line to the curve $y = x^2 + bx + c$ at the point $(9, -47)$.

a Find the derivative $\frac{dy}{dx}$ of $y = x^2 + bx + c$.

b State the gradient of the tangent to the curve at $x = 9$.

c Find the value of b.

d Find the value of c.

29 The curve $y = ax^3 + bx^2 + 2x - 17$ has a gradient of 58 at the point $(2, 31)$.

a Use the given information to write two equations expressing b in terms of a.

b Hence, find the value of a and b.

30 The graph of $y = ax^3 + bx^2 + cx + d$ has the following properties:

- It intersects the y-axis at $y = -28$, where the tangent is parallel to the x-axis.
- It intersects the x-axis at $(2, 0)$, where it has a gradient of 36.

a Use the information given in the first point above to find the value of c and d.

b Use the information given in the second point above to write two equations expressing b in terms of a.

c Hence, find the value of a and b.

31 From an external point $(3, 2)$, two tangents L_1 and L_2 are drawn to the curve $y = x^2 - 6$.

a Let the gradient of a tangent line to the curve be m. Find the two possible values of m that correspond to the gradients of L_1 and L_2.

b If L_1 is the tangent with the larger gradient, find the equation of L_1.

c Find the equation of L_2.

Normals

32 Consider the function $y = 10x^5 - 2x^4 + 8x^3 - 6x^2 + 4x - 164$.

a Find $\frac{dy}{dx}$.

b Find the gradient of the tangent to the curve at the point $(3, 2278)$.

c Find the gradient of the normal to the curve at the point $(3, 2278)$.

33 The gradient of the curve $f(x)$ at $x = 3$ is 2.

a Find the gradient of the tangent line at $x = 3$.

b Find the gradient of the normal line at $x = 3$.

34 Consider the curve $f(x) = x^2 + 8x + 15$ at the point $(4, 63)$.

a Find $f'(x)$.

b Find the gradient of the tangent to the curve at the point.

c Find the equation of the tangent to the curve at the point.

d Find the gradient of the normal to the curve at the point.

e Find the equation of the normal to the curve at the point.

35 Consider the curve $f(x) = 4x + \frac{64}{x}$ at the point $(4, 32)$.

a Find the equation of the tangent to the curve at the point.

b Find the equation of the normal to the curve at the point.

36 $-4x + y + 1 = 0$ is the normal to the curve $y = x^2 + bx + c$ at the point $(-8, -32)$.

a Find the derivative $\frac{dy}{dx}$ of $y = x^2 + bx + c$.

b State the gradient of the normal to the curve at $x = -8$.

c Find the value of b.

d Find the value of c.

Applications

37 Consider the curve $f(x) = 36 - x^2$ at the point $P(3, 27)$.

- **a** Find $f'(x)$.
- **b** Find the gradient of the tangent to the curve at the point.
- **c** Find the equation of the tangent to the curve at the point.
- **d** Find the gradient of the normal to the curve at the point.
- **e** Find the equation of the normal to the curve at the point.
- **f** The tangent line at $(3, 27)$ meets the x-axis at point A. Find the x-coordinate of A.
- **g** The normal line at $(3, 27)$ meets the x-axis at point B. Find the x-coordinate of B.
- **h** Find the length of AB.
- **i** Find the exact area of $\triangle ABP$.

38 The normal to the curve $y = x(x-5)^2$ at the point $A(6, 6)$ cuts the x-axis at B.

- **a** Find the gradient function.
- **b** Find the gradient of the tangent at A.
- **c** Find the gradient of the normal at A.
- **d** Hence, find the x-coordinate of point B.
- **e** Find the area of triangle whose vertices are the origin, point A and point B.